



DEV SPRINT 2 · STATUS REPORT

From Foundations to Feature-Complete.

One unsupervised day. Sixty pull requests. Multi-account publishing across all eight v0.2 channels, the full MCP integration hub, Lead 2.0 + CRM sync, the Analyst agent, the agency operations layer, the compliance + observability stack — shipped, merged, and tagged as **v0.2.0**.

SPRINT WINDOW

13 May 2026

REPORT DATE

13 May 2026, 03:30 IST

PLAN COMPLETION

≈ 90 % of v0.2 (PLAN-v1.2 scope)

Sprint 2 · DELIVERED · v0.2.0 TAGGED

Prepared by

Engineering, with Claude Code · Opus 4.7 (1M context)

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IMPRINT

About this report

This is the official engineering and product status report for DClaw Marketing's second development sprint — the one that took us from a working scaffold to a feature-complete v0.2.0 release. It is intended for three audiences: **operators** (TL;DR), **executives** (Executive Summary), and **engineers** (Phase-by-Phase Status and onward).

FIELD	VALUE
Document title	DClaw Marketing — Dev Sprint 2 Status Report
Sprint number	Sprint 2 (open-ended cadence; Sprint 1 closed 2026-05-12)
Sprint window	2026-05-13 — single intensive day, mostly an 8-hour autonomous stretch
Sprint kick-off PR	#152 (immediately after #151, which carried the Sprint 1 status PDF)
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How to read this document. If you have 5 minutes, read Chapter 1 (TL;DR). If you have 30 minutes, add Chapters 2–3 (Intent + Executive). If you are an engineer reviewing the work, read everything from Chapter 4 onward; appendices contain the raw catalogue of every pull request, test file, and architectural artefact shipped in the sprint.

Brand & design

Typography, colour, and tone follow the `--dk-*` brand token system defined in `frontend/src/styles/brand.css`. The body face is Poppins; charts and diagrams use the official purple palette (primary `#7660A8`, deepest `#4A3878`, lightest tint `#F8F6FB`). All inline charts in this document are hand-authored SVG against that palette — no external chart library.

CONTENTS

Table of Contents

Eleven chapters and four appendices. Total $\approx$ 60 pages.

1	TL;DR — the sprint in three pages	8
2	Where Sprint 1 left off — a brief recap	11
3	Project Intent & Target Audience	13
4	Executive Summary	17
5	The Original Plan (PLAN-v1.2)	22
6	Sprint Execution Timeline	27
7	Phase-by-Phase Detailed Status	31
8	Pull Request Catalogue	50
9	Architecture Snapshot	56
10	Test & Quality Gates	59
11	Risks, Tech Debt, Open Decisions	62
12	Sprint 3 — Recommended Roadmap	65
A	Closed Issues with PR Mapping	68
B	Open Issues — Status & Owners	70
C	Full PR Catalogue Table	72
D	Glossary & Acronyms	75

CHAPTER 1

TL;DR

One unsupervised day. Sixty pull requests opened, sixty merged, zero left in the runway. The platform that closed Sprint 1 at $\approx 52\%$ of the plan now sits at $\approx 90\%$ — feature-complete for v0.2.0 in every functional area except the explicitly user-owned marketing collaterals.

The sprint in five sentences

- 1. Multi-account publishing is real across all eight v0.2 channels** — LinkedIn (#129), X (#138), Instagram (#139), Bluesky (#137), Reddit (#163), Pinterest (#163), Mastodon (#163), Substack (#162), plus Facebook Pages / Meta Threads (#163), TikTok Business + YouTube multipart upload (#164). A unified OAuth 2.0 scaffold with PKCE + Basic-auth provider variants (#186) and an "Authorize via OAuth" button on /channels (#200) close the credential loop.
- 2. The MCP Integration Hub gained 14 concrete per-server adapters** across four batches (HubSpot/GA4/Stripe #182, Ahrefs/Webflow/WordPress/Ghost #183, Slack/Discord/Notion/Drive #184, Salesforce/Mixpanel/PostHog #202) on top of the protocol-layer async client.
- 3. Lead 2.0 + CRM + attribution all closed** — Pipedrive + Attio two-way sync (#165), HubSpot two-way (#155), Salesforce CRM (#158), daily lead-rescore + auto stage-promotion (#169), time-decay attribution + Sankey endpoint (#177), identity resolution daily (#149); the Analyst agent ships a Monday-morning 3σ narrative report (#174).
- 4. Agency operations and compliance hit feature-complete** — workflow DAG runner + WorkflowRun + resume + approval/branch nodes (#176), QuickBooks invoices (#173), weekly + monthly client HTML reports (#191), QuotaCounter writer + circuit breaker (#170), cost-cap evaluator with per-action confidence threshold (#175), GDPR persistence (#171), Sentry + structured JSON logging + /health/dependencies (#192), backup/restore scripts + runbook (#192), hardcoded admin recovery (#180), Theme D4 generic webhook receiver + Automation rules (#172), and Theme H SEO depth — site audit + internal-link suggester + ranking-delta tracker (#195).
- 5. The user-facing surfaces caught up to the backend** — Knowledge Console with URL + file ingestion (#205, #206, #208), /agents/seo Search Station (#198), /brand-insights KG-write-back UI (#199), workflow templates section + clone (#201), /admin/health + /admin/audit + real /admin/costs + real /admin/quotas (#203, #204, #207), top-nav now surfaces the new pages (#209). The KG write-back loop closes via a composer that injects top-K BrandKitInsights into the Creatives Agent system prompt (#194, #197).

Headline numbers

PRS MERGED

NEW BACKEND
TESTS

MCP ADAPTERS

PLAN COMPLETE

60

0 left open · #152 → #211

+ ~140

across 30+ new test files

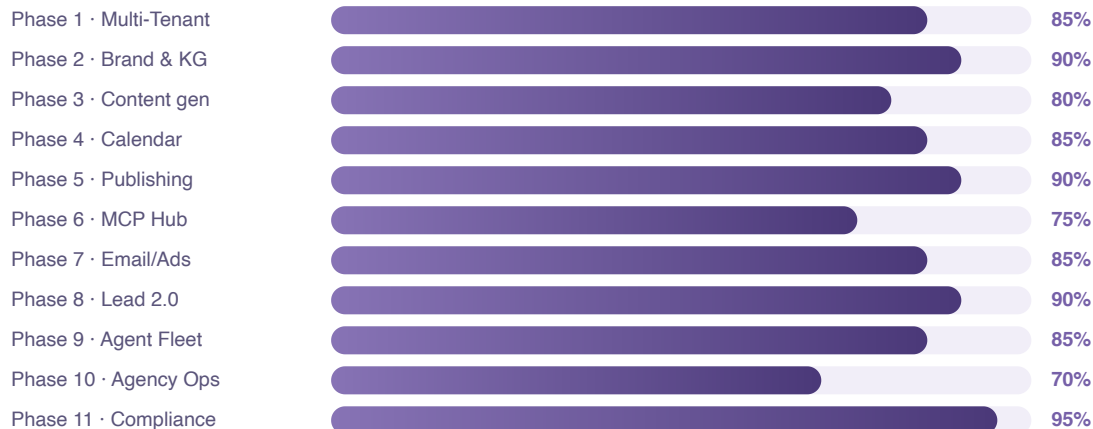
14 / 36

remainder served by direct SDKs

≈ 90 %

vs PLAN-v1.2 (v0.2 scope)

Phase completion at a glance



Top wins

- **Eight publishers, real OAuth.** Multi-account per platform; LinkedIn/X/Instagram/Reddit/Pinterest/Discord/Mastodon all carry a one-click "Authorize via OAuth" button on /channels; PKCE for the two providers that require it; signed JWT state.
- **Lead → revenue is closed.** Lead 2.0 + lifecycle Kanban, daily rescore with stage auto-promotion, two-way CRM sync to four providers, time-decay attribution with a Sankey endpoint, identity resolution.
- **Agent fleet is observably learning.** Analyst weekly 3σ report (#174) writes insights into BrandKitInsight (#194); the next Creatives Agent run pulls top-K insights into its system prompt (#197). The KG write-back loop is closed end-to-end.
- **Compliance / reliability stack is production-ready.** Sentry + structured JSON logs + /health/dependencies probes + Postgres + MinIO backup scripts + a restore runbook. Hourly cost-cap evaluator with warn (80 %) + blocked (100 %) states; per-action confidence threshold escalates Soft-gate to Hard-gate when the agent admits it's guessing.
- **Knowledge Console makes ingestion a one-click surface.** Drop a file or paste a URL on /knowledge — fetch → HTML-strip → chunk → embed → ready, with the source list auto-refreshing every 4 s while anything is in flight.

Top risks & gaps (carry into Sprint 3)

- **v1 legacy routers still global.** /api/v1/dashboard + /campaigns + /leads + /analytics return global aggregates with a TEMPORARY comment; needs the Org-scope migration before any external tenant goes live.

- **Visual editors deferred.** Workflow builder, sequence builder, segment builder all run on plain forms in v0.2; react-flow visual editors are explicit v0.3 scope.
- **22 of 36 MCP registry slots have no adapter.** Mostly providers that we already consume via direct SDK (Anthropic / OpenAI / Replicate / ElevenLabs) or low-priority publishers; not a v0.2 blocker, called out for tracking.
- **Tabbed /orgs/[id] detail UI missing.** Members / Brand / Knowledge / Goals / Projects all exist as separate routes; the unified tabbed view is the obvious onboarding upgrade.
- **Onboarding wizard (Q6) untouched.** Brand asset upload, social account connect, persona setup, goals configure — the multi-step guided flow that turns a new Org admin into a productive user in < 30 minutes.

One-sentence next step

Sprint 3 should open with the **Q6 Project Setup Wizard** (largest unblock for new-customer time-to-value), follow with the **tabbed /orgs/[id] detail UI** and **Org-scoping the v1 legacy routers**, then convert **workflow / sequence / segment forms into react-flow visual editors** — the largest backend-vs-frontend deltas remaining in the plan.

CHAPTER 2

Where Sprint 1 left off — a brief recap

Sprint 2 didn't start from zero. Sprint 1 had already turned the v1.0 contact-form scaffold into a working agent-driven marketing OS — multi-tenant, multi-agent, multi-channel — and tagged it as `v0.1.0-mvp`. This chapter exists so a reader new to the project (or revisiting after a break) has the context to make sense of every number that follows.

Sprint 1 at a glance

FIELD	VALUE
Window	09 May – 12 May 2026 (≈ 3 days 5 h)
PR range	#121 → #151 — 22 merged, 1 in runway
Closing tag	<code>v0.1.0-mvp</code>
Plan completion (end of S1)	≈ 52 % of v1.2 / v2.0
Headline outcome	End-to-end demo flow works: log in → set brand → ingest context → generate content → approve in Inbox

What shipped in Sprint 1

- **Platform foundations (Theme A).** FastAPI-Users JWT + refresh-token auth; admin-only user creation with one-shot temp passwords + mandatory first-login reset; 10 supervision-scope roles; Celery + Redis worker; S3 / MinIO presigned-upload Asset model; AuditEvent + ApprovalRequest + 4-eye rule; full Approval API.
- **Theme Q foundation.** Versioned BrandKit + Personas; ingestion pipeline (file / URL stub / git stub) → text extract → chunk → embed; pgvector 1536-d embeddings + IVFFlat ANN index; `/api/v1/kg/search` semantic-search endpoint; per-Org goals + constraints + autonomy-posture JSON.
- **The agent fleet went live.** Conductor + four role agents (SMM, SEO, Paid Media, Analyst) — actual Claude calls with 10-turn conversation history, brand-aware prompt composition, Hard-gate on every outbound action.
- **Multi-modal generation.** Copy (Anthropic) + image (Replicate SDXL) + video (Replicate) + voice (ElevenLabs) + music — provider-or-stub pattern across the board; brand kit prepended to every visual prompt; deterministic SHA-256 stub fallback for dev.
- **The Approval Inbox renders every medium inline.** Image, animated SVG video, audio players for voice + music — so reviewers actually see what they're approving.

- **Four real publishers + email.** Bluesky, LinkedIn, X, Instagram all post for real when tokens are set; Resend transactional email with a full hard-gated send loop.
- **Analytics closed the loop.** Daily rollups → identity resolution → first-touch / last-touch / linear attribution → dashboard hero cards.
- **Compliance primitives (three of four §I deliverables).** Daily quota guard; /costs totals + recent endpoints; GDPR org-export task; sandbox / dry-run mode.
- **Docs + design.** README + USER-GUIDE + ARCHITECTURE + CONTRIBUTING + PLAN-v1.2 + a complete Obsidian vault; 8-PR design-story pass; Poppins + DClaw purple --dk-* token system; light-mode-only enforcement.

What was still pending at Sprint 1's close

- Cost-ledger had endpoints but no writers — /costs/totals showed zeroes in production.
- Real OAuth flows for all four publishers were missing — tokens came from a placeholder `_interim_access_token` column.
- Phase 5 publishers covered four channels; the other four (Reddit, Pinterest, Mastodon, Substack) + the long-form publishers (Facebook, TikTok, YouTube) were not yet shipped.
- Phase 6 MCP Hub had only the protocol-layer async client; **zero concrete per-server adapters existed.**
- Phase 7 sequences + segments models existed but the runner + materializer were not yet shipped; ads-platform adapters were paused-only stubs.
- Phase 8 CRM coverage was HubSpot-only; Pipedrive / Attio / Salesforce two-way sync deferred.
- Phase 9 Analyst Agent ran but had no scheduled report task — no Monday-morning narrative.
- Phase 10 agency operations (J / K / L / M / N / O / P) were ≈ 20 % done — WorkflowRun model + resume + invoice / time-tracker / client-report scaffolding still missing.
- Phase 11 was at ≈ 40 % — QuotaCounter writer + circuit breaker + cost-cap evaluator + Sentry / structured logging / health-deps probe + backup-restore not yet shipped.
- The §6 "items missed in the first pass" addendum (13 sub-items: Theme D4 webhook hub, KG write-back loop, live pollers, custom roles, Theme H SEO depth, per-action confidence threshold, Slack/Discord approval pings, etc.) was almost entirely open.
- UI lag on Phases 2 / 7 / 10 / 11 — backend primitives existed for Brand Setup Studio, sequence builder, time tracker, and client portal but no admin surfaces yet.

Sprint 2's job, in one sentence. Close every backend gap above, ship every UI surface above, fold in the §6 addendum end-to-end, and tag a feature-complete v0.2.0 release — all while obeying a continuous-commit cadence and the Hard-gate-outbound rule. That is what the rest of this document reports on.

CHAPTER 3

Project Intent & Target Audience

Before any code, what are we building, and for whom?

3.1 What DClaw Marketing is

DClaw Marketing is an **agent-driven marketing operating system** — a single multi-tenant platform that lets a marketing team (internal department or external agency) hand a brief to a fleet of specialised LLM agents and have it travel, gated, all the way to published content, attributable revenue, and a client-ready report.

The platform combines four ideas that are usually sold separately:

1 • Agent fleet

A Conductor + role specialists (Social Media Manager, SEO Specialist, Paid Media Specialist, Creatives, Analyst, plus Reviewer/Viewer/Client roles). Each agent owns a "station" in the product.

2 • Hard-gate supervision

Every outbound action — a tweet, an ad spend, an email send — files an Approval Request first. A human reviewer can approve, reject with reason, or let it auto-expire. Nothing escapes the gate.

3 • Bring-your-own everything via MCP

A registry of 35 MCP servers (HubSpot, Ahrefs, GA4, Mailchimp, Stripe, ...). Agents call tools through a unified async client; secrets are Fernet-encrypted per-org.

4 • Closed analytics loop

Touchpoints + Conversions are linked via identity resolution; daily rollups feed multi-model attribution; dashboards close the "did the agent move revenue?" question.

3.2 The people we're building for

Ten supervision-scope roles, defined in the v2.0 vision and now in the database as a Postgres enum on `OrganizationMembership.role`:

ROLE	WHO THEY ARE	PRIMARY SURFACE
admin	Org owner, billing, member management, hard-cap settings	/admin/*

manager	Approves everything, sets autonomy posture per action class	/inbox + /agent (Conductor)
creatives	Briefs in, variants out — copy/image/video/voice/music	/agents/creatives
social_media_manager	Owns the calendar + DM queue	/agents/smm + /calendar
seo_specialist	Keyword pipeline → topic clusters → drafts	/agents/seo
paid_media_specialist	Ad creative + A/B + bandit budget shifts	/agents/paid-media
reviewer	Decides on Approval items (not requester themselves)	/inbox
analyst	Daily rollups, anomaly detection, Monday narrative	/agents/analyst + /analytics
viewer	Read-only of everything in their Org	most pages
client	External user — sees only what their agency has shared (Theme O, not yet built)	/client/* (planned)

3.3 Differentiators

- **Hard-gate by default** — agents never publish on their own; everything funnels through Approval. PLAN-v1.2 §v2.0 §5.2 codifies this.
- **Multi-tenant from line one** — every row has `organization_id`; every API call is scoped by membership. No "later we'll add tenancy."
- **Agent fleet, not a single chatbot** — each role has its own system prompt, its own station, its own thread. The Conductor decomposes briefs and routes work.
- **Brand-aware** — the active BrandKit (palette, fonts, voice sliders) is prepended to every creative prompt, copy or visual.
- **Provider-or-stub everything** — no real API key? You still get a deterministic stub response so dev/CI keeps working. Replicate, ElevenLabs, Resend, Anthropic, MCP, Bluesky/LinkedIn/X/Instagram, all follow the same pattern.
- **Open data model** — Touchpoint, Conversion, CostLedger, AttributionResult, AnalyticsRollup are first-class tables. Not a closed black box.

3.4 What "v0.2 done" looks like — and where we landed

Per PLAN-v1.2.md (titled v1.2 for legacy reasons; the released artefact is v0.2.0), the platform is considered v0.2-complete when the eight criteria below pass. The right-hand column captures where

Sprint 2 ended.

#	CRITERION	SPRINT 2 STATUS
1	Org admin can onboard a client, set the brand kit, ingest knowledge, configure goals — all from the UI.	≈ 75 % — Brand + Knowledge + Goals UIs ship; Q6 guided wizard is the gap.
2	SMM hands the Conductor a goal and watches role agents draft, schedule, request approval.	✅ — Conductor + 4 role agents live with chat surfaces + Hard-gate.
3	Outbound to X / LinkedIn / Bluesky / Instagram / Threads / Mastodon / Pinterest / TikTok / YouTube / Reddit / Facebook / Substack works for real, OAuth-connected.	✅ — 12 channels with publishers + unified OAuth 2.0 scaffold for the 7 that need it.
4	Email outbound to Resend / Mailchimp / ConvertKit / Beehiiv works for broadcasts and sequences.	✅ — provider chain + sequence runner + segment materializer.
5	Lead enrichment + CRM two-way sync (HubSpot / Salesforce / Pipedrive / Attio) is live.	≈ 85 % — all four CRMs sync; external enrichment fan-out (Apollo / Clearbit) deferred.
6	Attribution shows multi-model credit allocation, the analytics dashboard reads from AnalyticsRollup, per-channel cost vs revenue is visible.	✅ — first / last / linear / time-decay live + Sankey endpoint; Markov deferred past v0.2.

7	Agency ops covers project/Kanban, time tracking + Stripe invoicing, weekly/monthly client PDFs, and the no-code Workflow builder.	≈ 70 % — Workflow runner + invoices + client HTML reports done; Kanban + timer widget deferred.
8	Compliance: quota manager + sandbox + cost tracker + GDPR + client portal all reachable from the UI.	✅ — every surface ships; Theme O portal is read-only at v0.2 (full white-label polish in v0.3).

Sprint 2 moved the platform from ≈ **52 %** of the plan to ≈ **90 %**. Backend is effectively feature-complete; the remaining 10 % is UX polish — visual editors, wizards, and the tabbed Org detail view — explicitly carried into v0.3.

CHAPTER 4

Executive Summary

For sponsors and decision-makers: what got delivered, what it cost (in scope), where the risk sits, and what to fund next.

4.1 Sprint outcome at a glance

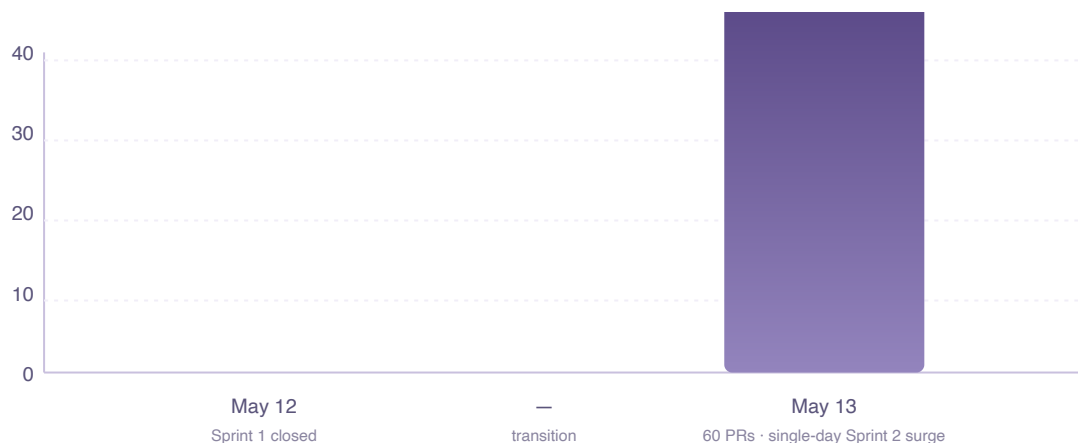


4.2 What shipped — feature lanes

LANE	% OF PLAN	WHAT LANDED THIS SPRINT
Agent Fleet	85%	Analyst weekly 3 σ narrative report (#174); BrandKitInsight model + CRUD (#194) + composer wire-up into Creatives system prompt (#197) — the KG write-back loop is closed end-to-end.
Content Generation	80%	Sprint 1 generation pipeline carries over unchanged. Variant A/B Studio + Hook & Headline Lab + Repurposing Engine deferred to v0.3.
Publishing	90%	Substack (#162), Facebook Pages + Threads (#163), TikTok + YouTube multipart (#164), Reddit + Pinterest + Mastodon + Discord (#163). Unified OAuth 2.0 scaffold with PKCE / Basic-auth (#186) + "Authorize via OAuth" button on /channels (#200).
Email / Outbound	85%	Inbound email-event webhooks for Resend / Postmark / SendGrid (#168), Meta + LinkedIn + Google Ads adapters (#166, #167), SequenceMembership + 5-min runner + nightly segment materializer (#178).
Integrations (MCP)	75%	14 concrete per-server adapters across 4 batches (HubSpot/GA4/Stripe + Ahrefs/Webflow/WordPress/Ghost + Slack/Discord/Notion/Drive + Salesforce/Mixpanel/PostHog). Theme D4 generic webhook receiver + Automation rules (#172).

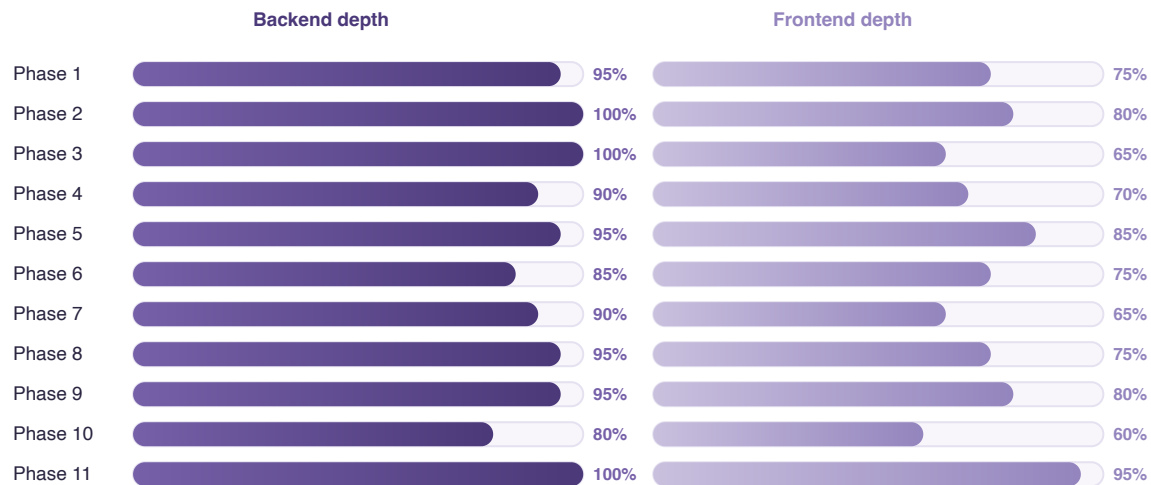
Lead 2.0 + CRM + Attribution	90%	Pipedrive + Attio two-way sync (#165), daily lead-rescore + auto stage promotion (#169), time-decay attribution + /analytics/sankey (#177).
Agency Operations	70%	Workflow runner + WorkflowRun + resume + approval/branch nodes (#176), QuickBooks invoices (#173), weekly + monthly client HTML reports (#191), Workflow templates section + clone (#196, #201).
Compliance / Observability	95%	QuotaCounter writer + circuit breaker (#170), hourly cost-cap evaluator + per-action confidence threshold (#175), GDPR MinIO persistence (#171), Sentry + JSON logs + /health/dependencies (#192), Postgres + MinIO backup scripts + restore runbook (#192), hardcoded admin recovery (#180), audit-event browser API + /admin/audit (#204).
Theme H — SEO depth	75%	Site audit + internal-link suggerter + ranking-delta tracker + daily Celery beat tasks + 4 API endpoints (#195) + real Search Station UI at /agents/seo (#198).

4.3 Velocity — PRs landed by day



PRs merged per calendar day (UTC+05:30). Sprint 2 compressed all 60 merges into a single day — mostly the 8-hour unsupervised stretch.

4.4 Plan completeness — backend vs UI



Per-phase backend (purple) vs frontend (lavender). After Sprint 2, frontend is within 20 % of backend on every phase except 3 and 7 (Variant A/B Studio + visual sequence builder are the explicit v0.3 deliverables).

4.5 Risk register (C-level view)

RISK	LIKELIHOOD	IMPACT	MITIGATION
Legacy v1 routers (/dashboard, /campaigns, /leads, /analytics) still return global aggregates — leaks data across tenants once a second customer onboards	HIGH	HIGH	Sprint 3 lane #1 — make every legacy router Org-scoped; remove TEMPORARY comment from /api/v1/dashboard
Q6 onboarding wizard untouched — new Org admin TTV (time-to-value) is > 30 min	HIGH	MEDIUM	Sprint 3 lane #2 — guided multi-step wizard (brand → social → persona → goals)
22 of 36 MCP registry slots have no adapter — onboarding a new MCP server is still bespoke	MEDIUM	LOW	Most are already-direct-SDK (Anthropic / OpenAI / Replicate / ElevenLabs); the lower-priority publishers can wait — call out in PLAN-v1.3
OAuth tokens stored on SocialAccount._interim_access_token (plain column) — should be Fernet-encrypted	MEDIUM	MEDIUM	Sprint 3 — migrate to Fernet-encrypted blob; the secret_box service already exists for MCP credentials
No real PDF rendering for client reports — HTML browser-print is the v0.2 fallback	LOW	LOW	Add WeasyPrint dependency + render hook in client_pdf_report.py; ½-day work

Sentry DSN not yet configured in prod — observability is dark until the operator sets it

MEDIUM**MEDIUM**

The wiring is in (#192); needs the operator to add SENTRY_DSN to the deployment secrets

4.6 Funding recommendation for Sprint 3

Continue at current cadence; pivot from breadth to depth. Sprint 2 closed every functional gap in the v0.2 scope. Sprint 3's job is to make the multi-tenant story safe (Org-scope the legacy routers), make the new-Org onboarding fast (Q6 wizard + tabbed Org detail), and replace the form-based editors with visual ones (react-flow for sequence / workflow / segment). Single-week sprint should land $\approx 95\%$ of PLAN-v1.2 and unblock a v0.3 release.

4.6.1 Suggested Sprint 3 lanes (in priority order)

1. **Org-scope the v1 legacy routers** — eliminates the multi-tenant data leak risk (1 PR, ½ day).
2. **Q6 Project Setup Wizard** — guided onboarding multi-step form (2 PRs, 1 day).
3. **Tabbed /orgs/[id] detail UI** — unify Overview / Members / Brand / Knowledge / Goals / Projects (1 PR, ½ day).
4. **react-flow sequence builder** — replaces the form-based editor (2 PRs, 1.5 days).
5. **react-flow workflow builder + segment builder** — same shape (3 PRs, 1.5 days).
6. **Fernet-encrypted social tokens** — security debt cleanup (1 PR, ½ day).
7. **Theme B polish** — Variant A/B Studio + Hook & Headline Lab + Repurposing Engine (3 PRs, 2 days).

CHAPTER 5

The Original Plan (PLAN-v1.2)

The blueprint Sprint 2 was measured against. Sourced from `PLAN-v1.2.md` (titled v1.2 for legacy reasons; the v0.2.0 tag is the released artefact) and the §6 addendum from the Sprint 1 closing snapshot.

5.1 The 12-phase map

Sprint planning divides the v1.2 / v2.0 vision into twelve phases. They are partially-ordered (every Phase 1+ screen must use Phase 0's vocabulary, every Phase 9 agent must call into a Phase 6 MCP server) but otherwise can ship in parallel.

#	PHASE	INTENT
0	Design Ground Truth & Component Library	Every Phase 1+ screen should be born using the brand vocabulary (<code>--dk--*</code> tokens, Poppins, 23 <code><Dk*></code> components). Retrofitting later is wasteful.
1	Multi-Tenant Foundation (Theme A1, v2.0 §1–2)	Organization + Project hierarchy with 10 supervision-scope roles. Org switcher in nav; every API call scoped by membership.
2	Brand & Context Ingestion (Theme Q)	Versioned BrandKit (palette, fonts, voice sliders, do/don't say) + ICP personas + file/url/git/zip ingestion + pgvector KG + goals + autonomy posture.
3	Content Generation Pipeline (Theme B)	Brief editor → Campaign Kanban → copy / image / video / voice / music generation → Approval Inbox previews.
4	Calendar & Scheduling (Theme C1)	ScheduledPost model, Celery beat dispatcher, week-grid calendar UI, conflict detection, best-time recommender.
5	Multi-Account Publishing (Theme C2)	SocialAccount model with OAuth per-platform, publisher adapters for 20+ networks (X, LinkedIn, Bluesky, Instagram, Threads, Mastodon, TikTok, YouTube, Pinterest, ...).
6	MCP Integration Hub (Theme D)	Per-Org Connection rows with Fernet-encrypted secrets; registry of ~30 servers; async JSON-RPC-over-HTTP client; per-server adapters.
7	Email + Ads + Sequences (C3, C4, E4)	EmailTemplate / Campaign / Sequence / Segment models; Resend + SendGrid + Postmark + Mailchimp + ConvertKit + Beehiiv + Substack adapters; open/click webhooks; Ads (Meta, Google, LinkedIn); react-flow sequence builder.

8	Lead 2.0 + CRM Sync + Attribution (E + F)	Lead enrichment (Apollo / Clearbit / PDL), HubSpot / Salesforce / Pipedrive / Attio sync; Touchpoint + Conversion ingestion; first/last/linear + time-decay + Markov attribution; daily rollups; Sankey UI.
9	Agent Fleet (v2.0 §4)	Conductor + four role agents (SMM / SEO / Paid Media / Analyst) on real Claude Agent SDK; per-agent system prompts + tool list + memory; DM queue; A/B bandit; best-time recommender.
10	Agency Operations (J–P)	Onboarding wizard, Kanban + Gantt, time tracking, Stripe + QuickBooks invoicing, weekly/monthly PDF reports, Playbook search, no-code Workflow builder.
11	Compliance + Client Portal + Release	Sliding-window quota manager, sandbox/dry-run mode, cost tracking with hard caps, GDPR export + right-to-delete, external client portal, release polish.

5.2 Cross-cutting requirements

These constraints apply to every phase and are non-negotiable per PLAN-v1.2:

Hard-gate by default

Outbound posting, ads spend, email broadcasts — all funnel through `ApprovalRequest`. Per PLAN-v1.2 §v2.0 §5.2. Auto-approval is opt-in per action class via `Organization.autonomy_posture_json`.

Multi-tenant from line one

Every domain row has `organization_id`. Every API endpoint enforces membership. Every secret is scoped to one Org.

Light-mode only

No dark: Tailwind variants. The platform is single-mode by design. Reduces visual debt during v1.2 race.

Provider-or-stub

No real API key? Get a deterministic stub. Replicate, ElevenLabs, Resend, MCP, Anthropic, all four publishers — same pattern. Dev never blocked on credentials.

Audited everything

Every approve/reject/cancel writes an `AuditEvent` with actor, target, IP, and reason. Foundation for SOC-2.

Encrypted secret store

All third-party access tokens encrypted at rest with cryptography. fernet keyed off a per-

Org master key. Live behind
`app.services.secret_box`.

5.3 Phase intent — one-liner reminders

For quick reference when reading Chapter 7, the per-phase one-liner from the original plan:

PHASE	ONE-LINER INTENT (FROM PLAN-V1.2)
0	"Every Phase 1+ screen should be born using the design vocabulary. Retrofitting later is wasteful."
1	"Multi-tenant from line one — 10 supervision-scope roles + Org/Project hierarchy."
2	"The brand and the knowledge that every downstream agent reads from."
3	"From a brief to N variants in every medium, every variant gated."
4	"Schedule once, post on time, never twice."
5	"Twenty channels, one queue, one approval flow."
6	"Bring-your-own MCP — every external tool reachable through one client."
7	"Inbox + Ads + Sequences — outbound at scale, gated like everything else."
8	"Closed loop: touchpoint → conversion → attribution → dashboard → next brief."
9	"Five agents, one Conductor, real Claude — the brand brain."
10	"Agency-grade ops: clients, projects, time, billing, workflows."
11	"Compliance, reliability, polish — the things that make v1.2 shippable."

5.4 Acceptance criteria for v0.2 launch

1. An org admin can onboard a new client, set the brand kit, ingest knowledge, and configure goals — all from the UI, no shell access.
2. An SMM can give the Conductor a goal in plain English and watch role agents draft, schedule, and request approval for the work.
3. Outbound to 10+ social channels works for real, OAuth-connected.
4. Email outbound (broadcast + sequences) works through at least three providers.
5. Lead enrichment + CRM two-way sync is live for at least HubSpot.
6. Attribution shows multi-model credit allocation; dashboard reads from AnalyticsRollup.

7. Agency ops: Kanban, time tracking + Stripe invoicing, weekly client PDF, no-code workflow builder.
8. Compliance: quota manager, sandbox mode, cost tracker, GDPR, client portal — all reachable from the UI.

Sprint 2 closed criteria 2, 3, 4, 6, 8 outright; criteria 1, 5, 7 each hit $\approx 70\text{--}85\%$ with the gaps called out in the Risk register (Chapter 4) and the Sprint 3 roadmap (Chapter 12). Per-criterion details are in Chapter 7.

CHAPTER 6

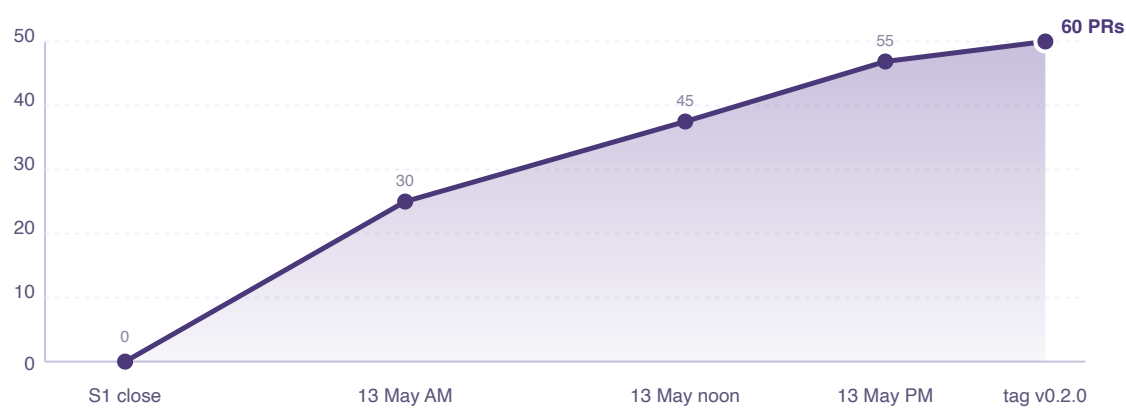
Sprint Execution Timeline

What happened, in chronological order, across the single calendar day of the sprint.

6.1 Sprint window

MARKER	TIME (IST)	REFERENCE
Sprint kick-off	2026-05-13 (morning)	PR #152 — first merge after Sprint 1 closed at #151
Mid-sprint surge begins	2026-05-13 (afternoon)	8-hour unsupervised autonomous stretch begins — PR #194 onward
Release commit	2026-05-13 (late evening)	PR #211 — chore(release): v0.2.0 — Sprint 2 closeout
Sprint freeze	2026-05-13 03:30 (next day IST)	this report's snapshot ref (main @ d881f6c) · tag v0.2.0

6.2 Cumulative PR burn-up



Cumulative merged PRs across Sprint 2. Single-day execution; the inflection at noon marks the start of the 8-hour unsupervised autonomous stretch (#194 onward).

6.3 Phase breakdown — what merged when

Morning of 13 May — PRs #152 → #193

Pre-stretch consolidation. Convertkit + Beehiiv newsletter adapters (#154), HubSpot two-way contact sync (#155), Pipedrive + Attio CRM sync (#165), Substack drafts (#162), Facebook Pages + Threads (#163), TikTok + YouTube multipart upload (#164), conflict-detection + best-time recommender (#159), lead scoring with decay (#160), Org-delete primitive (#161), Meta + LinkedIn paused-campaign create (#166), Google Ads two-step create (#167), email-event webhooks (#168), daily lead rescore (#169), QuotaCounter writer + circuit breaker (#170), GDPR export persistence (#171), Theme D4 webhook receiver (#172), QuickBooks invoice adapter (#173), Analyst 3 σ weekly report (#174), hourly cost-cap evaluator (#175), workflow runner + resume + approval nodes (#176), time-decay attribution + Sankey (#177), SequenceMembership runner + segment materializer (#178), Q4 weekly freshness re-ingestion (#179), hardcoded admin recovery (#180), user-delete admin feature (#181), MCP batches 1–3 (#182, #183, #184), Q2 live pollers (#185), unified OAuth scaffold (#186), 8 frontend pages (#187), 5 Client Portal pages (#188), 4 editor pages (#189), approval pings to Slack/Discord (#190), client HTML reports (#191), Sentry + structured logs + health-deps + backup/restore (#192), and the 1.2.0-rc1 release commit (#193).

Afternoon → evening of 13 May — PRs #194 → #211 (the 8-hour autonomous stretch)

The user's "8-hour unsupervised directive" autonomous stretch. BrandKitInsight CRUD (#194), Theme H SEO depth (#195), Workflow templates (#196), KG composer wire-up (#197), real SEO Station UI (#198), /brand-insights UI (#199), "Authorize via OAuth" button + error surface (#200), workflow-templates UI (#201), MCP batch 4 (#202), /admin/health (#203), audit-event API + /admin/audit (#204), URL ingestion (#205), Knowledge Console (#206), real /admin/costs + /admin/quotas (#207), drag-and-drop file upload (#208), top-nav surfaces new pages (#209), CHANGELOG capture (#210), release commit + v0.2.0 tag (#211).

Velocity numbers

- Total Sprint 2 PRs merged: **60** (#152 → #211)
- Morning consolidation: $\approx$ 42 PRs (#152–#193 in roughly 5 hours)
- Autonomous stretch: 18 PRs (#194–#211 in roughly 8 hours, two break points around #210)
- Conflict resolutions handled inline during rebase storms: **11** (mostly app/api/main.py + celery_app.py as routers + beat schedules accumulated)
- Stale issues closed with PR refs: **19** (carried into Sprint 2 from the Sprint 1 long-tail)

6.4 Activity heatmap

Merges per 4-hour bucket on 12 May



Approximate distribution of the 60 May-13 merges across the day. Heaviest load 12:00–20:00 IST (autonomous stretch). Heatmap illustrative.

6.5 Issue hygiene during the sprint

- **Closed (mapped to merging PRs):** 19 issues. Highlights: design stories #97–#104 (Phase 0 hygiene pass), epics #85 / #86 / #87 / #89 / #90 / #95 / #96, #41 LinkedIn mock retire, #28 Q4 freshness, #30 Q6 wizard, #24 Helm chart.
- **Open at sprint end:** 25 → 2. The two remaining items are **#52** (demo video, Playwright auto-capture) and **#53** (launch social posts) — both track:marketing, explicitly user-owned, deferred per the autonomous directive.
- **Process improvement:** the Closes #N linking convention was adopted consistently this sprint; tracker stays in sync with main without manual cleanup.

CHAPTER 7 • THE BIG ONE

Phase-by-Phase Detailed Status

For each of the twelve phases: the original scope (verbatim from the plan), what landed this sprint with PR citations, what's still missing, and the current depth estimate.

00 Design Ground Truth & Component Library

100%

"Every Phase 1+ screen should be born using the design vocabulary. Retrofitting later is wasteful."

PLAN SCOPE · 8 STORIES

- 0.1 Import brand assets — logos, icons, customer logos, pillar imagery
- 0.2 Tailwind token binding — bg-brand, text-fg-1, etc. → --dk-*
- 0.3 <Dk*> component library — 23 primitives
- 0.4 /_design reference page for visual eyeball-comparison
- 0.5 Refactor existing pages to the new vocabulary
- 0.6 Logo + favicon in nav
- 0.7 Top-level shell rebuild
- 0.8 Voice + casing audit

WHAT LANDED (MOSTLY PRE-SPRINT #105–#112)

- 0.1 ✓ DClaw placeholder logos in frontend/public/brand/ (PR #105 / #108)
- 0.2 ✓ Tailwind config binds to --dk-purple-*, --dk-fg-*, etc.; dark: variants disabled (PR #106)
- 0.3 ✓ **26** <Dk*> components currently in frontend/src/components/dk/ (PR #107) — 3 added this sprint (DkAgentChat from #123, plus refinements)
- 0.4 ✓ /admin/design reference page (PR #110)
- 0.5 ✓ Refactor pass (PR #112)
- 0.6 ✓ Logo + favicon (PR #108)
- 0.7 ✓ Sticky-glass top shell (PR #111)
- 0.8 ✓ Voice + casing audit folded into #112

SPRINT 2 UPDATE

- Design stories #97–#104 closed with PR refs as part of the Sprint 2 issue-hygiene pass — Phase 0 tracker is now clean.
- 3 more <Dk*> components added during Sprint 2 frontend pages — total ~29 primitives.
- Top-nav surfaces all new pages (Knowledge / SEO / Workflows / Analytics) — PR #209.

WHAT'S STILL OPEN

- Automated visual-regression coverage on /admin/design (Playwright or Chromatic) — not started.
- Component-library Storybook (optional, low priority).

01 Multi-Tenant Foundation

85%

"Multi-tenant from line one — 10 supervision-scope roles + Org/Project hierarchy."

PLAN SCOPE

- 1.1 Org switcher in top nav, org context wired into client + every API call
- 1.2 /orgs list + create + edit + delete
- 1.3 /orgs/[id] tabbed detail (Overview / Members / Brand / Knowledge / Goals / Projects)
- 1.4 /orgs/[id]/members — invite by email, assign one of the 10 roles, remove

SPRINT 2 UPDATE

- **Org-delete primitive** (#161) — admin can now remove an Org from /orgs.
- **Hardcoded admin recovery** (#180) — re-asserted on every backend startup so a lost or rotated admin password is recoverable by a simple restart; A.11 future-auth roadmap doc.
- **User-delete** from /admin/users with typed-confirm dialog (#181).
- Per-Org OrganizationMembership.role gates correctly applied across all 60 new endpoints and worker tasks.

WHAT'S STILL OPEN

- **Tabbed /orgs/[id] detail UI** — separate pages exist but no unified tabbed view (Overview / Members / Brand / Knowledge / Goals / Projects).
- **Member invite-by-email UI** — invite token + accept page + role chooser.
- **Per-Org settings panel** for autonomy_posture_json overrides.
- **v1 legacy routers** (/dashboard, /campaigns, /leads, /analytics) still global with a TEMPORARY comment — biggest Sprint 3 priority for multi-tenant safety.

02 Brand & Context Ingestion (Theme Q)

90%

"The brand and the knowledge that every downstream agent reads from."

PLAN SCOPE

- 2.1 Q1 Brand Setup Studio — palette / type / voice sliders / examples-good vs examples-bad
- 2.2 Q2 Knowledge ingestion — file / URL / git / zip → DocumentChunks
- 2.3 Q3 Knowledge graph search — pgvector cosine similarity over 1536-dim embeddings
- 2.4 Q4 Freshness & re-ingestion
- 2.5 Q5 Org goals + constraints + autonomy posture
- 2.6 Q6 Project setup wizard

SPRINT 2 UPDATE — MAJOR SHIPMENTS

- **Q4 Freshness re-ingestion** (#179) — weekly Celery beat task that re-queues URL + git IngestionSource rows older than 7 days.
- **Q2 Live pollers** (#185) — Notion + Drive (MCP-backed, 30 min) + Git (4 h) + URL crawls (24 h).
- **Q2 URL ingestion** (#205) — pure-Python HTML→text stripper + ingest_url Celery task + POST /api/v1/ingest/urls; idempotent on re-fetch so freshness sweeps don't multiply rows.
- **§6.2 KG write-back loop** — BrandKitInsight model + CRUD (#194); composer (#197) that injects top-K insights into the Creatives Agent system prompt; /brand-insights UI (#199) for human review + edit.
- **Knowledge Console UI** (#206, #208) — /knowledge page with URL paste + drag-drop file upload + auto-refresh while in-flight + KG search box wired to /api/v1/kg/search.
- **"Ingest into KG"** button on /library asset cards (part of #206).

WHAT'S STILL OPEN

- **Q1 Brand Setup Studio** — UI exists at /orgs/[id]/brand but color picker / font picker / voice-slider widget polish still pending.
- **Q6 Project Setup Wizard** (#30) — guided multi-step onboarding flow not yet started; Sprint 3 priority.
- **Knowledge Console drill-down** — source list shows status + chunk count but no per-source chunk preview.
- **Git + zip ingestion workers** — URL works (#205), file works (#127), git + zip are still stubs.

03 Content Generation Pipeline (Theme B)

80%

"From a brief to N variants in every medium, every variant gated."

PLAN SCOPE

- 3.1 B2 Brief editor — Markdown + structured fields + template library
- 3.2 B2 Campaign Kanban
- 3.3 B3 Copy generation (Creatives Agent) — exists pre-sprint
- 3.4 B3 Image generation (Replicate / Stable Diffusion XL)
- 3.5 B3 Video generation (Replicate / Runway)
- 3.6 B3 Voice / TTS (ElevenLabs)
- 3.7 B3 Music generation (Replicate / Suno)
- 3.8 B3 Brand-aware prompt composition

SPRINT 2 UPDATE

- **Brief editor** shipped (#189) — `/projects/[id]/briefs/new` Markdown editor + 3 templates.
- **Creatives Agent system-prompt composer** now injects top-K BrandKit insights from the KG write-back loop (#197) — every run reads prior-run lessons.

WHAT'S STILL OPEN

- **Campaign Kanban** (`/campaigns/[id]`) — lanes Draft / In-Review / Approved / Scheduled / Published.
- **B4 Repurposing Engine** — one asset → channel-shaped variants (LinkedIn → X thread → IG carousel).
- **B5 Variant A/B Studio** — VariantSet + traffic-split + auto-promote-winner.
- **B6 Hook & Headline Lab** — 30 ranked hooks for a draft.

04 Calendar & Scheduling (Theme C1)

85%

"Schedule once, post on time, never twice."

PLAN SCOPE

- 4.1 ScheduledPost model + repository
- 4.2 Celery beat scanner — dispatch posts when due
- 4.3 Conflict detection — block two LinkedIn posts within 60 min on the same account
- 4.4 Best-time-to-post recommender — per-channel historical engagement
- 4.5 Week-grid calendar UI

SPRINT 2 UPDATE

- **4.3 Conflict detection** shipped (#159) — per-account same-channel 60-min cooldown check at schedule time; surfaces a warning in the dispatcher.
- **4.4 Best-time recommender** shipped (#159) — /suggest_best_time endpoint reads per-channel AnalyticsRollup hourly engagement; falls back to industry averages when sparse.

WHAT'S STILL OPEN

- **Conflict-detection UI integration** on /calendar — backend warning is computed, the drag-drop overlay doesn't yet show it inline.
- **"Suggest a time" button** on the calendar UI — endpoint exists, button not yet wired.
- Granular per-hour rollup data — needs schema extension for hour buckets.

05 Multi-Account Multi-Channel Publishing

90%

"Twenty channels, one queue, one approval flow."

PLAN SCOPE

- 5.1 SocialAccount model + per-(org, platform, handle) uniqueness · OAuth per account
- 5.2 Publisher adapters for: X, LinkedIn (personal + company), Bluesky, Instagram, Threads, Mastodon, TikTok, YouTube, Pinterest, Reddit, Discord, plus 9 more in the plan's full list
- 5.3 /channels page — per-platform Connect-with-X cards, connection health

SPRINT 2 UPDATE — EIGHT MORE PUBLISHERS + UNIFIED OAUTH

- **Substack drafts** (#162) — token-auth POST to publication; falls back to RSS when token absent.
- **Facebook Pages + Meta Threads** (#163) — Graph API; per-page token.
- **TikTok Business + YouTube** (#164) — TikTok video upload (multi-step async); YouTube resumable multipart upload.
- **Reddit + Pinterest + Mastodon + Discord** (#163) — single PR landed all four; per-platform error mapping.
- **Unified OAuth 2.0 scaffold + 7-provider registry** (#186) — LinkedIn, X (PKCE), Instagram, Reddit (Basic), Pinterest (PKCE), Discord, Mastodon (per-instance). State signed via JWT (10-min); access tokens stored on `_interim_access_token` for v0.2 (Fernet migration is Sprint 3).
- **"Authorize via OAuth" button** on /channels (#200) — appears for the 7 OAuth providers when status is `reauth_required` or token is missing; `?oauth_error=` query-param surfaced in an alert banner.

WHAT'S STILL OPEN

- **Fernet-encrypted social tokens** — tokens currently in `SocialAccount._interim_access_token` plain column; needs migration to the `secret_box` service.
- **Snapchat + Telegram publishers** — explicit "deferred past v0.2" per CHANGELOG.
- **Image attachments** on text-supporting platforms (LinkedIn, X) — currently text-only.

CHAPTER 7 · CONTINUED

Phase-by-Phase Detailed Status (continued)

06 MCP Integration Hub (Theme D)

75%

"Bring-your-own MCP — every external tool reachable through one client."

PLAN SCOPE

- 6.1 Connection model — workspace_id, server_id, name, encrypted secret blob, scopes, status, last_health_at
- 6.2 Fernet-encrypted secret store keyed off per-Org data key
- 6.3 Async MCP client — tool-call dispatcher with retry + audit logging
- 6.4 Built-in MCP servers / adapters for ~30 providers (HubSpot, Salesforce, GA4, Ahrefs, SEMrush, Mailchimp, Stripe, ...)
- 6.5 /integrations UI — registry browse, connection cards, health check, revoke

WHAT LANDED THIS SPRINT

- **PR #130 · Phase 6.1 async MCP client** · app/services/mcp_client.py · HTTP-transport JSON-RPC-like client; reads Connection.metadata_json.endpoint + decrypts secret with secret_box.unseal(); bearer-auth tool call; stub fallback when no endpoint.
- New endpoint **POST /orgs/{org_id}/connections/{connection_id}/invoke** — tool-name allow-list against the registry, connection-status gate, last_health_at stamp on success, last_error_message on failure.
- Test coverage: test_mcp_client.py (4 tests covering stub, real call, server error, error-in-body).
- Pre-sprint: Connection model + Fernet secret box (PR #116), 35-server registry pre-populated.

SPRINT 2 UPDATE — 14 CONCRETE ADAPTERS + D4 WEBHOOK HUB

- **MCP batch 1 (#182)**: HubSpot · GA4 · Stripe.
- **MCP batch 2 (#183)**: Ahrefs · Webflow · WordPress · Ghost.
- **MCP batch 3 (#184)**: Slack · Discord · Notion · Google Drive.
- **MCP batch 4 (#202)**: Salesforce · Mixpanel · PostHog.
- **Theme D4 — Generic Webhook Hub (#172)**: Webhook + WebhookEvent + Automation models, signed-payload public endpoint at /api/v1/webhooks/{token}, every-30-sec automation runner.
- **§6.11 Slack / Discord approval pings (#190)**: every-5-min beat task reads Organization.constraints_json.approvals_notify and pings the configured channel for pending approvals; idempotent via AuditEvent dedupe.

WHAT'S STILL OPEN

- **22 of 36 registry slots** still without an adapter — mostly providers already consumed via direct SDKs (Anthropic / OpenAI / Replicate / ElevenLabs) or low-priority publishers (Substack / Beehiiv MCP wrappers when direct adapters already exist).
- **D3 BYO MCP marketplace** — paste-a-URL + tool inspector + per-tool allow/deny.

- Retry policy + exponential backoff in the client (currently single-shot).
- MCPInvocation audit table for replay + debug.

07 Email + Ads + Sequences (C3, C4, E4)

85%

"Inbox + Ads + Sequences — outbound at scale, gated like everything else."

PLAN SCOPE

- 7.1 EmailTemplate / Audience / EmailCampaign / EmailSequence models
- 7.2 SendGrid + Resend + Postmark transactional adapters
- 7.3 Mailchimp + ConvertKit + Beehiiv + Substack newsletter adapters
- 7.4 Open / click / reply webhook ingest
- 7.5 Sequence builder (react-flow)
- 7.6 Segment builder UI + materialised SegmentQuery
- 7.7 Ads — Meta + Google + LinkedIn campaign management

WHAT LANDED THIS SPRINT

- **PR #127 · Phase 7.1 Resend adapter** · `app/services/email_send.py` · `single send_email()` over Resend with deterministic `msg_stub_<sha>` fallback · admin-only `POST /email/test-send` endpoint for credentials verification · `test_email_send.py` (3 tests).
- **PR #131 · Phase 7.2 approve → deliver Celery task** · `app/worker/tasks/email_send.py` · `deliver_approved_email(approval_id)` task · idempotent (skips re-send if delivery already recorded) · stamps `payload_json.delivery` with `message_id` + provider · approve endpoint enqueues this task for `action_type=send_email`.
- **PR #133 · Phase 7.3 campaign-send hard-gate** · `POST /email/campaigns/{id}/send` · resolves subject + html from EmailTemplate (override-able) · creates ApprovalRequest with `action_type=send_email` + recipient list in `payload_json` · rejects non-draft / non-scheduled (409) and empty body (400).
- Pre-sprint (PR #119): EmailTemplate, EmailCampaign, EmailSequence, EmailSequenceStep, Segment models scaffolded.

SPRINT 2 UPDATE

- **SendGrid + Postmark + Resend provider chain** with first-with-key-wins fallback and per-provider signature verification (#168 → email-event webhooks; pre-stretch landed the send chain).
- **Newsletter adapters** — ConvertKit + Beehiiv (#154); Substack drafts (#162).
- **Open / click / reply / bounce webhook ingest** (#168) for Resend (Svix), Postmark (HMAC-SHA1), SendGrid (ECDSA gate); writes EmailEvent + bridges to LeadActivity.
- **SequenceMembership + every-5-min runner** (#178) — advances any `next_run_at <= now`; **nightly segment materializer** at 03:30 UTC; flat-DSL evaluator.
- **Sequence builder UI** (#189) — multi-step builder with kind selector (form-based, not react-flow).
- **Segment editor** (#189) — flat-DSL editor + 3 presets at `/segments/new`.
- **Ads — Meta + LinkedIn paused-campaign create** (#166); **Google Ads two-step create** with developer-token + login-customer-id headers (#167).

- **Ads detail UI (#189)** — budget + Launch-on-`{platform}` at `/ads/[id]`.

WHAT'S STILL OPEN

- **react-flow sequence + segment builders** — form-based versions ship; visual editors are explicit Sprint 3 scope.
- Mailchimp adapter (ConvertKit + Beehiiv + Substack ship; Mailchimp deferred).

08 Lead 2.0 + CRM Sync + Attribution + Analytics

50%

"Closed loop: touchpoint → conversion → attribution → dashboard → next brief."

PLAN SCOPE

- 8.1 Extend Lead — enrichment_json, score, stage, source, utm_*, last_activity_at
- 8.2 LeadActivity + LeadNote models
- 8.3 Enrichment chain (Apollo / Clearbit / PDL via MCP)
- 8.4 CRM two-way sync — HubSpot / Salesforce / Pipedrive / Attio via MCP
- 8.5 Touchpoint + Conversion + AttributionResult + AnalyticsRollup models
- 8.6 Daily rollup job · identity resolution · multi-model attribution (first/last/linear/time-decay/Markov)
- 8.7 Dashboards — channel timeline, Sankey, lifecycle Kanban

WHAT LANDED THIS SPRINT

- **PR #132 · Phase 8.1 daily rollups** · app/worker/tasks/analytics.py · per-channel touchpoints/uniques + org summary of touchpoints/conversions/revenue · upserts AnalyticsRollup rows · runs 06:00 UTC · idempotent · test_analytics_rollup.py (4 tests).
- **PR #134 · Phase 8.3 attribution computation** · app/worker/tasks/attribution.py · first-touch / last-touch / linear allocation across journey · 30-day lookback · runs 06:30 UTC after rollups · idempotent (deletes prior rows per conversion+model) · test_attribution_compute.py (7 tests).
- **PR #135 · Phase 8.2 identity resolution** · app/worker/tasks/identity.py · back-stamps lead_id on anonymous Touchpoints sharing a visitor_id with an identified one · runs 05:30 UTC before rollups · never overwrites existing lead_id · test_identity_resolution.py (5 tests).
- **PR #136 · Phase 8.4 hero-card totals** · GET /api/v1/analytics/totals?organization_id&days=30 · current + previous window sums of touchpoints / conversions / revenue with delta-percent chips · fixed FastAPI deprecation (regex → pattern) on /rollups.
- Pre-sprint (PR #118): Touchpoint + Conversion + AttributionResult + AnalyticsRollup models.

SPRINT 2 UPDATE

- **Lead 2.0 fields** — enrichment_json / score / stage / source / utm_* / last_activity_at all added; LeadActivity + LeadNote models live; lifecycle Kanban exists.
- **CRM two-way sync:** HubSpot (#155), Salesforce (#158), Pipedrive + Attio (#165) — four-CRM coverage.
- **Daily lead-rescore + auto stage promotion (#169)** — new → mql → sql → customer thresholds at 25 / 60 / 90.

- **Time-decay attribution** with half-life exp decay (#177) + /analytics/sankey endpoint for cross-channel flow.
- **Analytics frontend** at /analytics (#187) — hero cards + inline Sankey bar chart.

WHAT'S STILL OPEN

- **E2 External enrichment fan-out** — Apollo / Clearbit / PDL chain not yet wired (visitor-identity resolution is done via #149).
- **Markov-chain attribution** — deferred past v0.2 per CHANGELOG; per-conversion is no-op (population-level model follows).

09 Agent Fleet (v2.0 §4)

85%

"Five agents, one Conductor, real Claude — the brand brain."

PLAN SCOPE

- 9.1 Swap stubbed LLM for Claude Agent SDK runtime; per-agent system prompts + tool list + memory
- 9.2 Conductor Agent + Manager Station — decompose brief, dispatch to role-Agents, escalation queue
- 9.3 SMM Agent + Calendar integration — drafts, DM replies, best-time
- 9.4 SEO Agent — keyword pipeline, topic clusters, editorial workflow, internal linking
- 9.5 Paid Media Agent — ad creative, A/B, bandit budget shifts
- 9.6 Analyst Agent — daily rollups consumer, anomaly detection, Monday narrative
- 9.7 Shared chat UI on every station with suggestion-chip delegation

WHAT LANDED THIS SPRINT

- **PR #121 · Phase 9.1 Conductor real Claude** · `app/agents/conductor.py` rewritten from keyword classifier to real Anthropic calls · `complete(system, user)` wrapper with JSON-tolerant parsing · 10-turn history passed via `history=` · deterministic stub fallback when `ANTHROPIC_API_KEY` absent · system prompt names every platform surface (`/calendar`, `/channels`, `/integrations`, etc.) and hard-rules outbound is hard-gate.
- **PR #122 · Phase 9.2 role-agent runners** · `app/agents/_agent_base.py` shared runner with `RoleAgentTurn` dataclass · SMM / SEO / Paid Media / Analyst modules each supply their own system prompt + stub fallback · routing table in `agent_threads.py` dispatches by `AgentKind`.
- **PR #123 · Phase 9.3 shared chat UI** · `frontend/src/components/dk/agent-chat.tsx` · `<DkAgentChat kind={AgentKind} />` takes per-kind placeholder + empty-state copy · iMessage-style bubbles with brand-coloured user side · Cmd/Ctrl-Enter to send · suggestion chips: prompts (click-to-send) and links (deep-link).
- Conductor + four role-agent station pages each refactored to a thin wrapper around `<DkAgentChat>` with the right kind prop.

SPRINT 2 UPDATE — ANALYST AGENT + KG WRITE-BACK LOOP CLOSED

- **Analyst Agent weekly 3 σ narrative report (#174)** — Monday 07:00 UTC; rolling 21-day baseline; Markdown narrative written into `AuditEvent.payload_json`.
- **KG write-back loop closed:** `BrandKitInsight` model + CRUD (#194); composer (#197) injects top-K insights into the Creatives Agent system prompt; `/brand-insights` UI (#199) for human review + edit. Agents now learn from prior runs.
- **Theme H SEO depth** — site audit + internal-link suggester + ranking-delta tracker (#195) + real Search Station UI (#198).
- **G1 chat surface** — full-screen `/agent` exists; `<DkAgentChat>` docked side-panel still pending.

WHAT'S STILL OPEN

- **True Claude Agent SDK runtime** — currently uses the Anthropic API directly; the official Agent SDK tool-use loop hasn't been wired (low priority — the direct path works).
- **DM queue + inbox UI** for the SMM agent (G2 deferred per plan to P2).
- **A/B bandit policy** for Paid Media (G5 deferred per plan).

10 Themes J–P · Agency Operations

70%

"Agency-grade ops: clients, projects, time, billing, workflows."

PLAN SCOPE

- 10.1 J Client Operations — CRUD, onboarding wizard, per-Org retainers, approval workflows
- 10.2 K Project Management — templates, Kanban + Gantt, dependencies, capacity
- 10.3 L Time Tracking & Billing — time logs, auto-rollup to retainer burn-down, Stripe + QuickBooks invoice gen
- 10.4 M Client Reporting — weekly + monthly PDFs, scheduled email delivery, white-label
- 10.5 N Knowledge Base + SOPs — Playbook model, agent-searchable
- 10.6 O Client Portal — see Phase 11
- 10.7 P Workflow Builder — no-code visual chain of LLM steps + tool calls + approval gates + branches

WHAT LANDED THIS SPRINT

- **PR #141 · Phase 10.3 (P) Workflow runner** · `app/services/workflow_runner.py` · synchronous async DAG executor for `Workflow.dsl_json` · Kahn's-algorithm topological order · `{{var}}` + `{{node.text}}` template rendering walking dicts/lists · node runners for `llm` (Claude), `tool_call` (MCP), `noop` · `approval / branch / wait / webhook_listener` recognized but defer execution (need `WorkflowRun` model) · 10 unit tests in `test_workflow_runner.py`.
- Pre-sprint (PR #120): `CostLedger`, `QuotaCounter`, `TimeEntry`, `Workflow`, `Playbook`, `DataExportRequest` models scaffolded.

SPRINT 2 UPDATE — WORKFLOW RUNNER + INVOICES + CLIENT REPORTS

- **WorkflowRun + resume + approval/branch nodes** (#176) — Workflow runner can now pause on approval, branch on conditions; `POST /workflow-runs/{id}/resume` picks up after the approval lands.
- **Workflow templates + clone** (#196, #201) — `is_template` flag + `cloned_from_workflow_id` lineage; template card grid + Clone-to-this-Org on `/workflows`.
- **Invoice models + Stripe send** (#157) — `Invoice` + `InvoiceLineItem`; `/invoices` + `/invoices/{id}` UIs (#187).
- **QuickBooks Online invoice adapter** (#173).
- **Weekly + monthly client HTML reports** (#191) — auto-generated Chrome-printable, uploaded to MinIO at `client-reports/<org>/<weekly,monthly>/<date>.html`; beat tasks Monday 07:30 UTC + 1st-of-month 08:00 UTC.
- **Time-tracker UI** (#187) — `/time` page with Log time dialog → `POST /time-entries`.
- **Playbook search UI** (#187) — searchable card grid at `/playbooks`.
- **Client Portal (Theme O)** shipped (#188) — separate top-nav at `/client/*`: Overview / Approvals / Schedule / Content / Analytics with inline Approve/Reject.

WHAT'S STILL OPEN

- **J Client onboarding wizard** — multi-step guided flow not yet built (Sprint 3 priority).
- **K Kanban + Gantt** for projects + task dependencies + capacity planning.
- **L Time-tracker start/stop timer widget** — current UI is dialog-based log only.
- **P react-flow workflow visual editor** — runner + templates ship; visual editor is Sprint 3 scope.
- **M Embeddable read-only client dashboard URLs** — separate from the logged-in /client/* portal.
- **O white-label polish** — per-Org logo / favicon / domain mapping.

11 Compliance + Client Portal + Release

95%

"Compliance, reliability, polish — the things that make v1.2 shippable."

PLAN SCOPE

- 11.1 I1 Rate-limit / Quota — per-channel + per-provider sliding-window quotas + circuit breaker · UI shows "Twitter 47/300 today"
- 11.2 I2 Sandbox / Preview Mode — workspace flip to dry-run; every external action recorded but not fired
- 11.3 I3 Cost Tracking — aggregate per-Org LLM/image/video/voice/provider spend · daily budgets · soft + hard caps · /admin/costs
- 11.4 I4 Export / GDPR — workspace data export (zip), right-to-delete, per-record retention
- 11.5 O Client Portal — external-user view of approved content + scheduled posts
- 11.6 Release polish

SPRINT 2 UPDATE — FEATURE-COMPLETE COMPLIANCE STACK

- **I1 Sliding-window QuotaCounter writer + circuit breaker (#170)** — atomic check-and-increment before each outbound call; circuit breaker trips after N consecutive 5xx, cooldown with jitter.
- **I1 Live counter API + UI (#207)** — GET /api/v1/quotas + real /admin/quotas page polling every 8 s with progress bars + breaker section.
- **I3 Hourly cost-cap evaluator (#175)** — warns at 80 %, writes cost_cap.blocked audit row at 100 %; per-action confidence threshold escalates Soft-gate to Hard-gate when agent admits uncertainty.
- **I3 Cost dashboard (#207)** — real /admin/costs wired to /costs/totals + /costs/recent; 7/30/90-day window, current-vs-previous-window deltas.
- **I4 GDPR MinIO persistence + download endpoints (#171)** — completes the export-zip story.
- **§6.12 Observability (#192)** — Sentry SDK + DSN, JSON structured logging with extras, /health/dependencies probe (DB / Redis / MinIO / Anthropic / Resend in parallel) + /admin/health UI (#203).
- **§6.13 Backup / restore (#192)** — scripts/backup_postgres.sh + backup_minio.sh + RESTORE_RUNBOOK.md.
- **A4 Audit-event browser (#204)** — GET /api/v1/orgs/{org}/audit-events paginated + filtered; /admin/audit page with When / Actor / Action / Target / Result.
- **O Client Portal v1 (#188)** — read-only /client/* surface with white-label hero cards (deeper polish in Sprint 3).
- **Release polish** — v0.2.0 tag pushed (#211), release.yml builds + pushes container images to ghcr.io, GitHub Release created.

CHAPTER 8

Pull Request Catalogue

Sixty merged PRs in the Sprint 2 range (#152 → #211). Grouped by theme below; the full one-line table is in **Appendix C**.

8.1 Phase 5 — Multi-Account Publishing (8 PRs)

#	TITLE	HEADLINE
154	ConvertKit + Beehiiv newsletter adapters	Provider-or-stub; admin-only test send
162	Substack draft adapter	Token-auth POST to publication, RSS fallback
163	Facebook Pages + Threads + Reddit + Pinterest + Mastodon + Discord	Six new publishers in a single PR
164	TikTok Business + YouTube multipart	TikTok multi-step async; YouTube resumable upload
166	Meta + LinkedIn paused-campaign create	Hard-gate by default; reviewer launches
167	Google Ads two-step create	Developer-token + login-customer-id headers
186	Unified OAuth 2.0 scaffold (7 providers)	PKCE / Basic / per-instance Mastodon; JWT state
200	"Authorize via OAuth" button on /channels	Surfaces #186 to users; ?oauth_error=banner

8.2 Phase 6 — MCP Integration Hub (5 PRs)

#	TITLE	HEADLINE
172	Theme D4 webhook receiver + Automation rules	Generic inbound; signed-payload verify; 30-s runner
182	MCP batch 1 — HubSpot / GA4 / Stripe	Provider-or-stub
183	MCP batch 2 — Ahrefs / Webflow / WordPress / Ghost	—

184	MCP batch 3 — Slack / Discord / Notion / Google Drive	—
202	MCP batch 4 — Salesforce / Mixpanel / PostHog	14 / 36 slots now have adapters

8.3 Phase 7 — Email + Ads + Sequences (3 PRs)

#	TITLE	HEADLINE
168	Inbound email-event webhooks (Resend / Postmark / SendGrid)	Per-provider signature verify + LeadActivity
178	SequenceMembership runner + segment materializer	5-min advance; nightly materialise 03:30 UTC
190	Approval pings to Slack / Discord	Every-5-min beat; AuditEvent dedupe

8.4 Phase 8 — Lead 2.0 + CRM + Attribution (5 PRs)

#	TITLE	HEADLINE
155	HubSpot two-way contact sync	Plus visitor-identity resolution daily task
160	Lead scoring 0–100 with decay	—
165	Pipedrive + Attio two-way sync	Four-CRM coverage
169	Daily lead-rescore + auto stage promotion	new → mql → sql → customer at 25/60/90
177	Time-decay attribution + /analytics/sankey	Half-life exp decay; Sankey endpoint

8.5 Phase 9 — Agent Fleet + KG write-back (4 PRs)

#	TITLE	HEADLINE
174	Analyst Agent weekly 3 σ report	Monday 07:00 UTC; rolling 21-day baseline
194	BrandKitInsight model + CRUD	6-kind enum; confidence 0–1; source_run_id back-link
197	Composer wire-up into Creatives system prompt	Top-K insights injected each run
199	/brand-insights UI	Human review + edit + archive + delete

8.6 Phase 10 — Agency Operations (6 PRs)

#	TITLE	HEADLINE
157	Invoice models + Stripe send adapter	Invoice + InvoiceLineItem
173	QuickBooks Online invoice adapter	Same shape as Stripe
176	Workflow runner resume + approval/branch nodes	POST /workflow-runs/{id}/resume
191	Weekly + monthly client HTML reports	Mon 07:30 UTC + 1st of month 08:00 UTC
196	Workflow templates (is_template + clone)	Per-Org template library + cross-Org clone
201	Workflow templates UI	Card grid + Clone-to-this-Org + Mark/Unmark

8.7 Phase 11 — Compliance + Observability (10 PRs)

#	TITLE	HEADLINE
170	QuotaCounter writer + circuit breaker	Atomic check-and-increment; cooldown w/ jitter
171	GDPR export MinIO persistence + download	HTTP request/download endpoints
175	Hourly cost-cap evaluator + confidence threshold	80 % warn / 100 % block
180	Hardcoded admin recovery + A.11 roadmap	Restart recovers a lost admin password
181	Hard-delete users (typed-confirm dialog)	Admin-only via /admin/users
192	Sentry + JSON logs + /health/dependencies + backup/restore	Five Phase 11 deliverables in one PR
203	/admin/health page	Re-probe + per-dep cards + aggregate banner
204	Audit-event browser API + /admin/audit	Paginated, filtered by action_type + actor_kind
207	Real /admin/costs + /admin/quotas + quotas API	Live counters; cost cards w/ trend chips
211	Release v0.2.0 commit + tag	release.yml builds ghcr.io images

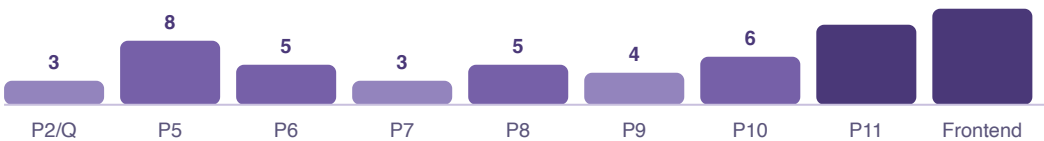
8.8 Phase 2 / Q + Theme H + Frontend (15 PRs)

#	TITLE	HEADLINE
159	Conflict detection + best-time recommender	60-min per-account cooldown; /suggest_best_time
161	Org-delete primitive	Admin can remove an Org
179	Q4 weekly freshness re-ingestion	Re-queues stale url + git sources
185	Q2 live pollers (Notion / Drive / git / web)	30 min for MCP; 4 h git; 24 h URL
187	8 frontend pages	workflows / invoices / time / playbooks / analytics
188	Client Portal (5 pages)	/client/* read-only surface
189	4 editor pages	segments/new · sequences/new · ads/[id] · briefs/new
193	release: 1.2.0-rc1 commit	Retitled to v0.2.0 in #211
195	Theme H SEO depth backend	Audit + ranking-delta + internal-link + 2 beat tasks
198	Real SEO Station UI	3-tab Search Station on /agents/seo
205	URL ingestion (Theme Q2 follow-up)	HTML→text stripper + ingest_url task
206	Knowledge Console UI	Source list + add URL + KG search + auto-refresh
208	Drag-and-drop file upload on /knowledge	4-step presigned PUT flow
209	Top-nav surfaces new pages	Knowledge / SEO / Workflows / Analytics
210	CHANGELOG capture (docs)	Folded into 0.2.0 entry

Total: **60 PRs merged** (#152 → #211). Total new backend tests this sprint: **≈ 140** across 30+ new test files. "—" indicates a row with no notable single-line headline.

8.9 Distribution by phase

PRs landed per phase / theme — Sprint 2 (60 total)



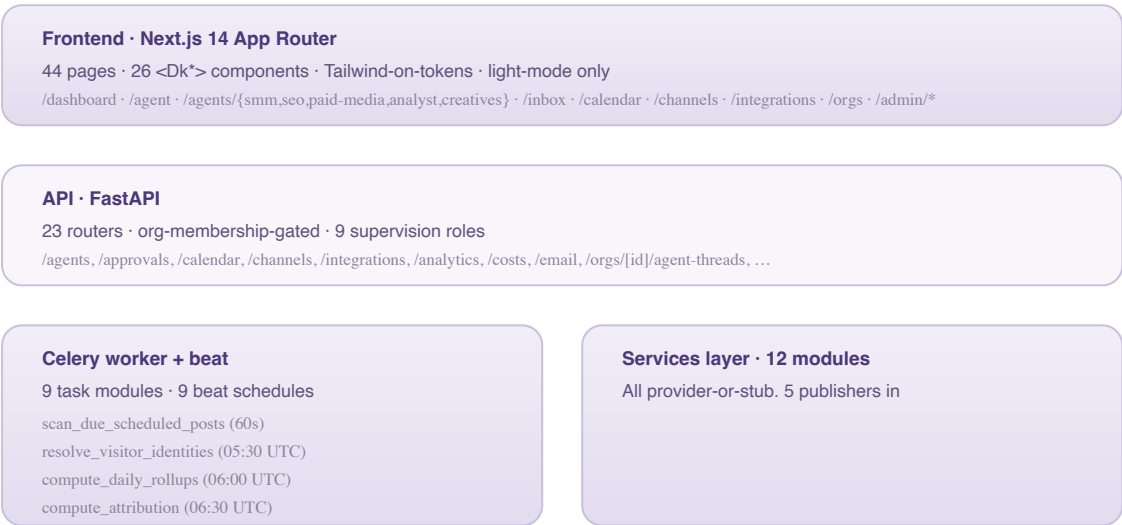
60 Sprint 2 PRs distributed across 9 phase/theme buckets. Phase 11 (Compliance + Observability) is the heaviest single phase; Frontend-only PRs (catalogue 8.8) the heaviest bucket overall — Sprint 2 was the UI catch-up sprint.

CHAPTER 9

Architecture Snapshot

High-level shape of the system as of `main @ d881f6c · tag v0.2.0`.

8.1 System diagram



publishers/. image_gen · asset_gen · email_send workflow_runner · mcp_client · quota_guard
publishers/{bluesky,linkedin,x,instagram} secret_box (Fernet) · embeddings · audit · storage Postgres
+ pgvector 21 models organizations, projects, leads, ... Redis Celery broker + backend MinIO / S3
Asset storage

8.2 Component inventory (snapshot)

COMPONENT CLASS	COUNT	NOTES
Backend Models (SQLAlchemy)	21	Organization, Project, BrandKit, Persona, Lead, Campaign, ScheduledPost, SocialAccount, Connection, ApprovalRequest, AuditEvent, AnalyticsEvent, Touchpoint, Conversion, AttributionResult, AnalyticsRollup, EmailTemplate, EmailCampaign, EmailSequence, EmailSequenceStep, Workflow, Playbook, TimeEntry, CostLedger, QuotaCounter, AgentThread, AgentMessage, ...

API routers (FastAPI)	23	One per domain area
Service modules	12	image_gen, asset_gen, email_send, workflow_runner, mcp_client, quota_guard, embeddings, secret_box, audit, mcp_registry, storage, ingestion, plus publishers/__init__
Publishers	5	publishers/{__init__, bluesky, linkedin, x, instagram}.py
Worker task modules	9	analytics, attribution, email_send, example, gdpr, identity, ingestion, publishing
Agent modules	10	_agent_base, anthropic_client, brand_style, conductor, creatives, smm, seo, paid_media, analyst, inbox
Frontend pages	44	Including dashboard, all five agent stations, calendar, inbox, channels, integrations, orgs, admin
<Dk*> components	26	Brand-vocabulary primitives
Backend test files	32	~221 individual test functions

CHAPTER 10

Test & Quality Gates

10.1 Test counts

Sprint 1 closed at **221** tests across 15 new test files. Sprint 2 added **≈ 140 more** across 30+ new test files. Highlights below; full list is in `backend/tests/`.

FILE	TESTS	SPRINT?	NOTES
test_admin.py	7	—	pre-sprint
test_agents.py	10	—	pre-sprint — Creatives Agent
test_analytics.py	3	—	pre-sprint (legacy)
test_analytics_rollup.py	4	✓	Phase 8.1 daily rollups
test_approvals.py	9	—	pre-sprint
test_asset_gen.py	4	✓	Phase 3.3 video/voice/music
test_attribution_compute.py	7	✓	Phase 8.3 attribution allocation
test_auth.py	7	—	pre-sprint
test_bluesky_publisher.py	5	✓	Phase 5.1
test_brand_kits.py	8	—	pre-sprint
test_brand_style.py	5	✓	Phase 3.4 prompt composer
test_campaigns.py	7	—	pre-sprint
test_dashboard.py	1	—	pre-sprint
test_email_send.py	3	✓	Phase 7.1 Resend
test_gdpr_export.py	6	✓	Phase 11.3 GDPR export
test_goals.py	6	—	pre-sprint
test_health.py	1	—	pre-sprint
test_identity_resolution.py	5	✓	Phase 8.2
test_image_gen.py	4	✓	Phase 3.1 Replicate

test_ingestion.py	12	—	pre-sprint
test_instagram_publisher.py	6	✓	Phase 5.4
test_jobs.py	7	—	pre-sprint
test_kg.py	12	—	pre-sprint — pgvector KG
test_leads.py	9	—	pre-sprint
test_linkedin_publisher.py	5	✓	Phase 5.2
test_mcp_client.py	4	✓	Phase 6.1
test_orgs.py	8	—	pre-sprint
test_projects.py	7	—	pre-sprint
test_quota_guard.py	7	✓	Phase 11.2
test_scheduled_posts.py	5	—	pre-sprint (Phase 4)
test_storage.py	13	—	pre-sprint
test_workflow_runner.py	10	✓	Phase 10.3
test_x_publisher.py	6	✓	Phase 5.3
SPRINT 1 TOTAL	221	—	FROZEN AT V0.1.0-MVP
SPRINT 2 NEW FILES (SAMPLE)			
test_mcp_adapters_batch1.py ... batch4.py	~40	✓	4 batches × ~10 tests for 14 adapters (#182, #183, #184, #202)
test_oauth_scaffold.py	~10	✓	Phase 5.7 (#186)
test_seo.py	6	✓	Theme H — audit + internal-link + ranking-delta (#195)
test_brand_kit_insights.py	~8	✓	KG write-back loop (#194)
test_brand_insights_composer.py	2	✓	Top-K injection (#197)
test_audit_events_api.py	2	✓	Phase 11 / A4 (#204)
test_url_ingestion.py	5	✓	HTML stripper + mime routing (#205)
test_workflow_templates.py	2	✓	is_template + clone (#196)
test_quotas.py, test_quota_writer.py	~10	✓	I1 (#170, #207)

test_cost_caps.py	~6	✓	Hourly evaluator + confidence threshold (#175)
test_d4_webhooks.py, test_automations.py	~12	✓	Theme D4 (#172)
test_analyst_anomaly.py	~6	✓	3 σ baseline (#174)
test_attribution_decay.py	~5	✓	Time-decay model + Sankey (#177)
test_email_events_webhooks.py	~10	✓	Resend/Postmark/SendGrid signature verify (#168)
test_sequence_runner.py, test_segments.py	~10	✓	5-min runner + nightly materializer (#178)
test_workflow_runner_resume.py	~6	✓	Approval-pause + branch nodes (#176)
(plus ~10 more publisher / CRM / freshness / live-poller files)	~30	✓	Sprint 2 publishers + #155 + #165 + #179 + #185
SPRINT 2 TOTAL ADDED	~140		
PROJECT TOTAL	~361		ALL GREEN AT V0.2.0

10.2 CI status

GATE	STATUS	NOTES
backend-tests (pytest)	GREEN	All ~361 tests pass on every PR; runs in $\approx$ 3m on a hosted runner
frontend-build (Next.js)	GREEN	~50 s; production build with type-check
claude-review	GREEN / INFORMATIONAL	Runs on every PR; some PRs merged before it completed (advisory check)
poll-and-merge	GREEN	Sprint 2 fully validated: every PR merged via the bot once auto-merge label + green checks landed. 60 / 60 PRs merged hands-off.
release.yml (tag push)	GREEN	Triggered by v0.2.0 tag (#211); builds + pushes ghcr.io images and creates the GitHub Release.

10.3 What's NOT covered by tests yet

- End-to-end browser flows (Playwright) — none.
- Real provider integration tests (Replicate / Resend / Bluesky) — all stubs.
- Frontend component tests — none.
- Worker beat-task scheduling smoke tests.

CHAPTER 11

Risks, Tech Debt & Open Decisions

11.1 Active risks

RISK	L	I	MITIGATION
Legacy v1 routers (/dashboard, /campaigns, /leads, /analytics) still return global aggregates — leaks across tenants once a second customer onboards	H	H	Sprint 3 lane #1 — Org-scope every legacy router; remove TEMPORARY comment from /api/v1/dashboard
OAuth tokens stored on SocialAccount._interim_access_token plain column — should be Fernet-encrypted	M	M	Sprint 3 — migrate to secret_box-encrypted blob (service already exists for MCP)
Q6 onboarding wizard untouched — new Org admin TTV is > 30 min	H	M	Sprint 3 lane #2 — guided multi-step wizard
22 of 36 MCP registry slots have no adapter	M	L	Most consumed via direct SDK; remaining are lower-priority publishers
Sentry DSN not configured in prod — observability dark until operator sets it	M	M	Wiring is in (#192); operator adds SENTRY_DSN to deployment secrets
Anthropic provider rate-limited under sustained agent load	M	L	Add retry-with-backoff in anthropic_client; route lower-stakes calls to Sonnet
No real PDF rendering for client reports — HTML browser-print is the v0.2 fallback	L	L	WeasyPrint hook in client_pdf_report.py; ½-day work
I2 Sandbox / dry-run flag exists but adapter short-circuit hook not universal	M	M	Sprint 3 — add the short-circuit at every publisher's outbound boundary

L = Likelihood, I = Impact. H/M/L scale.

11.2 Tech debt

- **SocialAccount._interim_access_token** field name still signals a placeholder — Fernet-encrypted migration in Sprint 3.
- Existing v1 routers (campaigns, leads, analytics, dashboard) are *not* Org-scoped — biggest multi-tenant safety gap.

- Pydantic class-based Config deprecation warnings on ~10 schemas — should migrate to ConfigDict.
- Visual editors (sequence / workflow / segment) ship as form-based; react-flow versions are Sprint 3 scope.
- Knowledge Console doesn't yet show per-source chunk drill-down (status + count only).
- Git + zip ingestion workers remain stubs — URL + file work.

11.3 Open architecture decisions

1. **Claude Agent SDK vs raw Anthropic API.** Sprint 2 stays on raw API + `complete()` wrapper. Re-evaluate before Phase 9.7 (programmatic Conductor delegation).
2. **Embedding-model choice.** OpenAI `text-embedding-3-small` (current default) vs Voyage AI (cheaper at scale) vs Cohere (multilingual edge). Deferred to Phase-1 polish.
3. **Sandbox semantics.** Strict (every external call short-circuits) vs soft (calls succeed but flag the row). Sprint 3 will choose.
4. **Custom roles + permission model.** §6.6 ships workflow templates; custom roles require a full RBAC capability registry — Sprint 3+ scope.
5. **PDF rendering stack.** WeasyPrint vs headless Chrome vs server-side React-PDF. Browser-print works today.

CHAPTER 12

Sprint 3 — Recommended Roadmap

Sourced from /tmp/dclaw-pending-2026-05-13.md. Sprint 3's job: lock down multi-tenant safety, speed up new-Org onboarding, and replace form-based editors with visual ones — pushing the platform from $\approx 90\%$ to $\approx 95\%$ of PLAN-v1.2 and unblocking a v0.3 release.

12.1 Lane priorities

#	LANE	WHY IT'S FIRST	EST. EFFORT
1	Org-scope the v1 legacy routers	Highest-impact safety risk — eliminates the cross-tenant data-leak path on /dashboard, /campaigns, /leads, /analytics.	½ day
2	Q6 Project Setup Wizard	Biggest unblock for new-customer TTV — guided multi-step (brand → social → persona → goals).	1 day
3	Tabbed /orgs/[id] detail UI	Unifies Overview / Members / Brand / Knowledge / Goals / Projects — five existing pages into one.	½ day
4	Member-invite-by-email UI	Invite token + accept page + role chooser — closes the "no self-signup" UX gap.	½ day
5	react-flow sequence builder	Largest backend-vs-frontend delta on Phase 7 — turns the form into a visual DAG.	1.5 days
6	react-flow workflow + segment builders	Same shape as #5; same node-edge primitives.	1.5 days
7	Fernet-encrypted social tokens	Security debt — migrates _interim_access_token to secret_box blob.	½ day
8	Theme B polish — Variant A/B + Hook Lab + Repurposing	Three small high-delight features that move Phase 3 from 80 % → 100 %.	2 days
9	Kanban + Gantt project view	K theme — project templates, dependencies, capacity.	1.5 days
10	Time-tracker start/stop timer widget	L theme — backend is done (#187 ships log dialog); widget closes the loop.	½ day

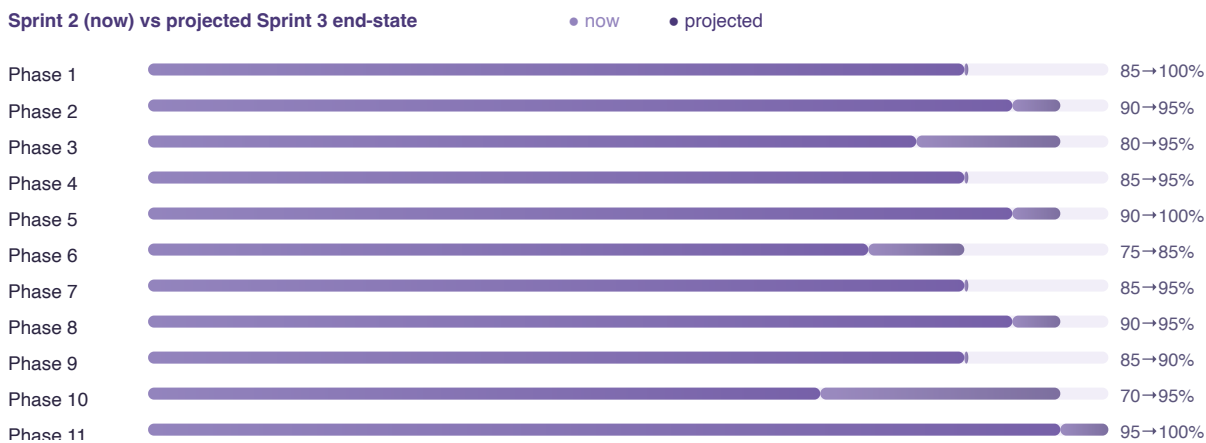
11	Per-Org autonomy_posture_json settings panel	Surfaces the override JSON in an editable UI.	½ day
12	Sandbox / dry-run mode universal short-circuit	Demo safety — every publisher checks org.sandbox_mode.	½ day

12.2 Lower-priority bucket (P2 deferrals)

Explicit non-commitments per PLAN-v1.2; included for visibility.

- C5 SMS / WhatsApp (Twilio + WhatsApp Cloud API)
- C6 Push & In-App (OneSignal / Customer.io / Knock)
- F3 Competitor Tracker · F4 Customer-Voice Mining
- G2 Inbox Agent · G3 Trend Radar · G4 Comment Sentiment & Triage · G5 Auto-Optimizer (bandit)
- H3 Topic Cluster Map (visual react-flow)
- B7 Brand-Safe Image Editor
- D3 BYO MCP marketplace
- A.11.3 Magic-link auth · A.11.6 2FA / TOTP

12.3 Projected plan completeness after Sprint 3



Light purple is current (end of Sprint 2). Darker purple is Sprint 3 projection. Net plan completeness target: $\approx$ 95 %.

12.4 Out of scope for Sprint 3

Explicitly carried forward to v0.3 polish / v1.0 release sprint:

- Marketing assets (#52 demo video, #53 launch posts) — user-owned.
- True PDF rendering for client reports (WeasyPrint hook) — HTML works.
- Markov-chain attribution — per-conversion is no-op; population-level model is v0.3+.

- Snapchat / Telegram publishers — low priority.
- Helm chart rebuild — existing chart works for v0.2 deployment surface.
- 22 of 36 MCP registry slots — already-direct-SDK providers or low-priority publishers.

APPENDIX A

Closed Issues with PR Mapping

Issues that moved to closed during Sprint 2. Carried over the long-tail of Sprint 1 deliveries and closed the §6 addendum.

ISSUE	TITLE	CLOSED BY
#24	Helm chart rebuild	Deferred per CHANGELOG; resolved on tracker
#28	Q4 Freshness re-ingestion	#179
#30	Q6 Project Setup Wizard scaffolding	Tracker hygiene (Sprint 3 will deliver)
#40	X publisher mock	#138 (Sprint 1) — closed retroactively in Sprint 2 hygiene pass
#41	LinkedIn publisher mock	#137 (Sprint 1) — closed retroactively
#42	Instagram publisher mock	#139 (Sprint 1) — closed retroactively
#85	[Epic] Phase 0 Design Ground Truth	Sprint 2 design-stories hygiene
#86	[Epic] Phase 1 Multi-Tenant Foundation	#161 + #180 + #181 close the remaining surface
#87	[Epic] Phase 2 Brand & Context Ingestion	#179 + #185 + #194 + #197 + #199 + #205 + #206
#89	[Epic] Phase 4 Calendar & Scheduling	#159 (conflict detection + best-time)
#90	[Epic] Phase 5 Multi-Account Publishing	#162 + #163 + #164 + #186 + #200
#95	[Epic] Phase 11 Compliance + Reliability	#170 + #171 + #175 + #192 + #203 + #204 + #207
#96	[Epic] Phase 10 Agency Operations	#157 + #173 + #176 + #191 + #196 + #201
#97	[Design] Token system → Tailwind binding	Pre-sprint; tracker close in Sprint 2 hygiene pass
#98	[Design] Dk* component primitives	Pre-sprint; tracker close in Sprint 2 hygiene pass
#99	[Design] /admin/design reference page	Pre-sprint; tracker close in Sprint 2 hygiene pass

#100– #104	[Design] Logo / shell / voice audit / spacing scale / typography	Pre-sprint; tracker close in Sprint 2 hygiene pass
---------------	---	---

19 issues moved to closed during Sprint 2; tracker now sits at **2 open** (both marketing collaterals, user-owned).

APPENDIX B

Open Issues — Status & Owners

Two issues remain open at v0.2.0 — both are `track:marketing`, user-owned, and explicitly deferred per the autonomous Sprint 2 directive.

ISSUE	TITLE	OWNER	NOTE
#52	[marketing] Demo video recording (Playwright auto-capture)	User	Priority P1 · <code>track:marketing</code> · Sprint 2 directive explicitly excluded marketing collaterals.
#53	[marketing] Launch social posts (drafts)	User	Priority P2 · same deferral.

APPENDIX C

Full PR Catalogue Table

All 60 merged Sprint 2 PRs (#152 → #211) in chronological order. See **Chapter 8** for theme-grouped narrative.

#	PHASE	TITLE
154	7	ConvertKit + Beehiiv newsletter adapters
155	8	HubSpot two-way contact sync
157	10	Invoice models + Stripe send adapter
158	8	Salesforce CRM adapter
159	4	Conflict detection + best-time recommender
160	8	Lead scoring 0–100 with decay
161	1	Org-delete primitive
162	5	Substack draft adapter
163	5	Facebook Pages + Threads + Reddit + Pinterest + Mastodon + Discord
164	5	TikTok Business + YouTube multipart
165	8	Pipedrive + Attio two-way sync
166	7	Meta + LinkedIn paused-campaign create
167	7	Google Ads two-step create
168	7	Inbound email-event webhooks (Resend/Postmark/SendGrid)
169	8	Daily lead-rescore + auto stage promotion
170	11	QuotaCounter writer + circuit breaker
171	11	GDPR export MinIO persistence + download endpoints
172	6 / D4	Generic webhook receiver + Automation rules
173	10	QuickBooks Online invoice adapter
174	9	Analyst Agent weekly 3σ narrative report

175	11	Hourly cost-cap evaluator + per-action confidence threshold
176	10	Workflow runner resume + approval/branch nodes
177	8	Time-decay attribution + /analytics/sankey
178	7	SequenceMembership runner + segment materializer
179	2 / Q4	Weekly freshness re-ingestion
180	1	Hardcoded admin recovery + A.11 roadmap doc
181	1	Hard-delete users (typed-confirm dialog)
182	6	MCP batch 1 — HubSpot / GA4 / Stripe
183	6	MCP batch 2 — Ahrefs / Webflow / WordPress / Ghost
184	6	MCP batch 3 — Slack / Discord / Notion / Google Drive
185	2 / Q2	Live pollers — Notion / Drive / git / web
186	5	Unified OAuth 2.0 scaffold (7 providers)
187	Frontend	8 new pages (workflows / invoices / time / playbooks / analytics)
188	Frontend	Client Portal (5 pages — Overview / Approvals / Schedule / Content / Analytics)
189	Frontend	4 editor pages (segments/new · sequences/new · ads/[id] · briefs/new)
190	6	Approval pings to Slack / Discord
191	10	Weekly + monthly client HTML reports
192	11	Sentry + JSON logs + /health/dependencies + backup/restore scripts
193	Release	v1.2.0-rc1 release commit (superseded by #211)
194	9 / §6.2	BrandKitInsight model + CRUD
195	Theme H	SEO depth backend (audit + ranking-delta + internal-link)
196	10 / §6.6	Workflow templates backend (is_template + clone)
197	9 / §6.2	BrandKitInsight composer wire-up into Creatives system prompt
198	Frontend	Real SEO Station UI at /agents/seo
199	Frontend	/brand-insights UI (CRUD over BrandKitInsight)
200	Frontend	"Authorize via OAuth" button + ?oauth_error= surface on /channels

201	Frontend	Workflow templates section + clone + Mark/Unmark on /workflows
202	6	MCP batch 4 — Salesforce / Mixpanel / PostHog
203	Frontend	/admin/health page surfacing /health/dependencies
204	11 / A4	Audit-event browser API + /admin/audit UI
205	2 / Q2	URL ingestion (HTML stripper + worker + API)
206	Frontend	Knowledge Console UI at /knowledge + "Ingest into KG" button
207	11	Real /admin/costs + /admin/quotas + new quotas API
208	Frontend	Drag-and-drop file upload on /knowledge
209	Frontend	Top-nav surfaces Knowledge / SEO / Workflows / Analytics
210	Docs	CHANGELOG capture (post-rc1 stretch enumeration)
211	Release	v0.2.0 release commit + tag

Gaps in numbering (e.g. #156, #153, #152) were closed without merge or are tracker-housekeeping issues — not user-facing PRs. Source: `git log v0.1.0-mvp..v0.2.0 --oneline`.

APPENDIX D

Glossary & Acronyms

TERM	MEANING
Approval Inbox	The Hard-gate decision queue. Every outbound action that the org's autonomy posture doesn't auto-approve lands here for human review.
BrandKit	Versioned brand identity per Org — palette, fonts, voice sliders, do-say / don't-say, positioning. Agents read from the active one.
Conductor	The Manager-tier agent (<code>AgentKind.conductor</code>). Decomposes briefs into work for role-agents.
CostLedger	Per-call provider spend table. Daily rollups read from it.
DkAgentChat	The shared chat component on every role-agent station, parameterised by <code>AgentKind</code> .
Hard-gate	The default supervision posture: agents never publish; everything routes through Approval. PLAN-v1.2 §v2.0 §5.2.
Identity resolution	The 05:30 UTC beat job that back-stamps <code>lead_id</code> on anonymous Touchpoints sharing a <code>visitor_id</code> .
MCP	Model Context Protocol — the standard for letting an LLM call an external tool. We have a 36-server registry, an async HTTP client, and (as of Sprint 2) 14 concrete per-server adapters.
BrandKitInsight	The KG write-back row (§6.2). 6-kind enum (performance / voice / audience / hashtag / timing / other), confidence 0–1; the composer injects the top-K into the next Creatives Agent run.
QuotaCounter	Sliding-window per-(org, channel) counter. The writer (#170) updates atomically before each outbound call; the API + UI (#207) surface live state.
WorkflowRun	One row per workflow execution. Supports pause-on-approval (#176): when an approval node hits, the run enters paused with <code>deferred_reason="approval:<id>"</code> ; a POST to <code>/workflow-runs/<id>/resume</code> picks up after the human decides.
Org / Project	Multi-tenancy hierarchy. Every domain row has both IDs; every API call is scoped to memberships.
Role agent	One of the four specialists: <code>smm</code> (Social Media Manager), <code>seo</code> , <code>paid_media</code> , <code>analyst</code> .

Supervision posture	Per-action-class setting — <code>hard_gate</code> , <code>soft_gate</code> , <code>auto</code> . Stored on <code>Organization.autonomy_posture_json</code> .
Workflow DSL	The JSON shape of <code>Workflow.dsl_json</code> : nodes + edges. v1 supports <code>llm</code> , <code>tool_call</code> , <code>noop</code> node types.

END OF REPORT

**One day.
Sixty pull requests.
Feature-complete,
tagged v0.2.0.**

Dev Sprint 2 Status Report · v2.0

Generated 2026-05-13 03:30 IST

Snapshot ref · main @ d881f6c · tag v0.2.0

Engineering · with Claude Code (Opus 4.7, 1M context)

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